STEM Website

Roadmap progress, backend hardening, verification evidence, and manual deployment gate

```
Repository: C:/Users/anath/OneDrive/Desktop/AI EXPO 2610/STEM Website
```

Scope: local repository only. No secrets were read or included. No new game modules were implemented.

Executive summary

The platform has moved from a functional prototype toward a controlled learning platform. Authentication, authoritative game validation, tutor fallback handling, CI, automated tests, legacy RPC cleanup, idempotent attempts, and dependency pinning are now implemented. The final direct-validator revoke is prepared but deliberately paused until the updated frontend is confirmed live and tested.

Verified evidence

• 32 automated Node tests pass.

• Repository verification passes.

• Supabase migration 012 was applied successfully; the idempotent RPCs have authenticated execute permission.

• Migration 013 is prepared locally but has not been applied.

• No Edge Function redeployment is required for attempt idempotency.

Completed implementation

Area	Completed work	Evidence
Authority	Removed the client fallback to record_game_attempt. Unsupported payloads fail without an RPC call.	js/progress-api.js; platform tests
Legacy access	Migration 011 revoked client access to legacy progression, achievement, and quota RPCs.	migrations/011_revoke_legacy_client_rpcs.sql
Authentication	PhysicsSim and RoboMaze adapters fail closed when authentication is missing or incomplete.	PhysicsSim/adapter.js; RoboMaze/adapter.js
Idempotency	Migration 012 added attempt_receipts and wrapper RPCs for retry-safe Physics and RoboMaze attempts.	migrations/012_attempt_idempotency.sql
Final cleanup	Migration 013 is ready to revoke direct client access to the old authoritative validator RPCs.	migrations/013_revoke_direct_authoritative_rpcs.sql
Dependencies	Browser Supabase CDN references are pinned to 2.118.0.	login.html; dashboard.html; both game pages
CI and tests	Deno npm resolution and Deno.test compatibility were fixed; regression coverage was added.	quality.yml; deno.json; tests/

Roadmap status

Phase	Status	Assessment
0. Protect state	Complete	Stabilization branch and preserved history.
1. Reproducibility	Mostly complete	README, config, CI, and pinned CDN exist. package-lock.json still requires a machine with npm.
2. Shared APIs and auth	Mostly complete	Shared APIs exist; some game orchestration remains inline.
3. Backend authority	Substantially complete	Server validation and reward authority exist. Final direct-validator revoke awaits deployment verification.
4. Game refactor	Partial	Adapters, levels, simulation, and UI files exist; entry pages remain large.
5. Server validation	Implemented, integration gap	Both validators exist; real database behavior tests remain.
6. Tutor reliability	Implemented	Retries, timeout, refund, states, and fallback exist. Provider 429/503 remains external.
7. Tests and checks	Partial	32 local tests and CI checks pass; browser E2E, accessibility, HTML, link, and DB tests remain.
8. Module contract	Partial	Contract exists, but persistence and lifecycle are not fully behind it.
9. Future games	Not started	Correctly deferred until the current platform passes release gates.

Why implementation is paused

Migration 012 intentionally retained the old authoritative RPC grants during a compatibility window. Migration 013 must only be applied after the updated frontend is live. Applying it sooner could break an older deployed frontend that still calls the original validator names.

Manual gate

The following actions require your Supabase and deployment access:

• Confirm the commit containing progress-api.js and migration 012 is pushed and GitHub Actions is green.

• Open the updated frontend.

• Complete one failed and one successful PhysicsSim attempt.

• Complete one failed and one successful RoboMaze attempt.

• Confirm level unlocking and rewards still work.

• Confirm attempt_receipts exists in Supabase.

After these checks, apply migration 013 in Supabase SQL Editor and rerun the privilege query.

Verification SQL

```
select to_regclass('public.attempt_receipts');

select p.proname, has_function_privilege('authenticated', p.oid, 'execute') as
authenticated_can_execute from pg_proc p join pg_namespace n on n.oid = p.pronamespace where
n.nspname = 'public' and p.proname in ('submit_robo_maze_attempt', 'submit_physics_attempt',
'submit_robo_maze_attempt_idempotent', 'submit_physics_attempt_idempotent') order by p.proname;
```

Remaining work after the manual gate

• Apply migration 013 after the updated frontend is verified, then confirm old validator privileges are false and wrapper privileges remain true.

• Generate package-lock.json using npm and change CI from npm install to npm ci.

• Add real Supabase integration tests for RLS, level locking, duplicate submissions, rewards, and quota behavior.

• Finish moving game persistence, access, reset, and server-result handling behind the module contract.

• Add browser E2E, accessibility, HTML, and link checks.

• Add privacy-safe tutor request IDs, provider status, latency, and terminal-state logs without prompts or student responses.

• Add CSP and replace unsafe inline handlers and dynamic HTML where practical.

• Define the final future-game schema and validation contract; do not add new game modules until the current games pass release gates.

Safety notes

• No service-role key, Gemini key, prompt, access token, or student response was included.

• No Edge Function redeployment or new game module was performed.

• The hosted frontend and database behavior must be confirmed manually before the final revoke migration is applied.